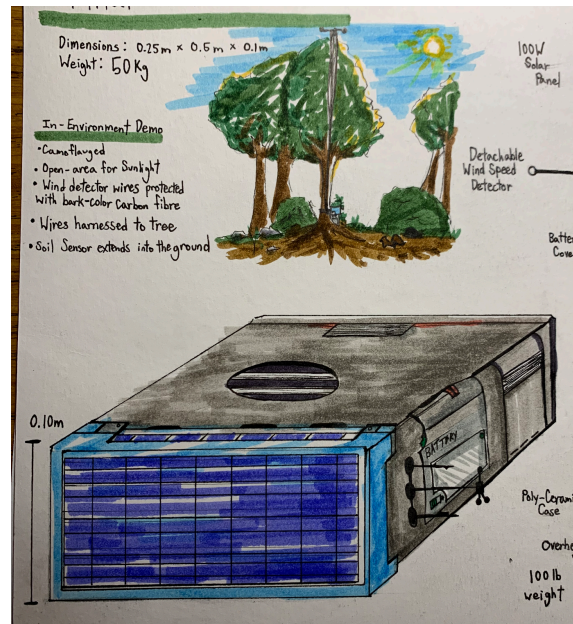


Smart Detection System for Early Wildfire Prevention



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Executive Summary

With the growing frequency and impact of wildfires, early detection systems and risk models are extremely important. To address the need for early wildfire detection, we have designed a device with the necessary environmental sensors and transmission capabilities to detect wildfires and generate data for wildfire risk assessment.

Problem: Lack of Quick Detection for Wildfires

Currently, wildfire detection relies primarily on satellite systems. These satellites are effective at tracking the growth of fires once they have been identified but struggle to detect fires in a timely manner, or small fires at all. For example, the Terra satellite used in NASA's Fire Information Resource Management System (FIRMS) makes two orbits every day, and has an infrared resolution of 1km x 1km, leaving many fires undetected [4].

Project and Solution Concept: Smart Detection System for Early Wildfire Prevention

This sensing system is a self-sustaining and weatherproof device that collects and transmits important environmental data for wildfire detection and risk analysis. This design includes temperature, air humidity, soil moisture, wind, and VOC gas sensors, all of which provide key insight into the risk of a wildfire starting in the area. The device is supplied with energy through solar panels that are mounted on the outside of the case, providing enough energy to power the continuous operation of the device. Additionally, the case material of fiberglass-infused plastic is weather resistant and has a high heat tolerance, allowing for the system to continue functioning through various weather conditions. Finally, the data is transmitted through a long-range antenna, which will send the data to a location for processing.

Project Requirements:

Based on tests and research, the device successfully meets all hardware requirements, including power consumption, weight, water resistance, fire resistance, and camouflage. The fire resistance of the selected case material, fiberglass-infused plastic, exceeds the required 800°C by 400°C. By formulating a table of performance validations, it is demonstrated that the system meets all criteria. The sensory accuracy is verified to have a worst-case error of $\pm 3\%$, well within the $\pm 5\%$ threshold. The data relay has a range of 20 km, and the wildfire detection speed is validated to respond well within 10 minutes. The total design was calculated at \$1,150.

Social Impact and Future Improvements

Although not quantifiable, the introduction of a ground-based detection device would be greatly beneficial given that it solves current systems' shortcomings. The greatest impact our device would have is the countless people and structures that would be spared from the destruction. Cost-wise, our device offers the trade-off of needing to place a device throughout fire-prone regions and potentially spending around 20 to 50 thousand USD in exchange for the increased protection of the public and wildfire predictability. This is not a significant concern as governments and large organizations are our primary users and have the budget required to operate the device. In addition, the device has a low environmental footprint being solar powered. Our device could potentially be made readily available to a global market for national government agencies around the world, and with proper adjustments to the cost and functionality, it could also be marketable to average consumers. The device may also benefit from further development to minimize its shortcomings such as the sensor accuracy, its size, or cost of production.

This report explains our device concept, project goals and objectives, project requirements, proof of concept, and results.

Introduction

Wildfires cause significant damage in the U.S. every year and are generally caused by human negligence. However, the inefficiency of current wildfire prevention systems and the absence of reliable detection tools often lead to further unnecessary damage to structures and the environment. More specifically, the lack of a ground-based detection system contributes greatly to this dilemma. To combat this problem, we propose a unique device capable of detecting wildfires through the implementation of multiple sensors and the capability to withstand extreme environmental conditions and relay data locally.

1.1 Background

A wildfire is an uncontrollable fire that incinerates forests, grasslands, and other dry high-density vegetation biomes. Wildfires can burn vegetation within and above the soil. Ground fires often start in soil rich with organic material, such as plant roots, which fuel the flames. These fires can smolder for extended periods, sometimes an entire season, until conditions enable them to develop into surface or crown fires. In contrast, surface fires burn dry or dead vegetation lying just above the ground, such as dry grass or fallen leaves. Crown fires, however, burn through the leaves and canopies of trees and shrubs [1].

These fires cause significant irreversible damage to structures as well as large areas of land in the United States despite efforts to combat them. In 2023 there were 56,580 reported wildfires in the United States, which burned over 2.5 million acres of land. That is more land than the entire state of Delaware. These fires also caused great damage to infrastructure, destroying 4,318 buildings, which included 3,000 homes [2]. Despite the common intuition that wildfires occur only in California, wildfires occur all over the United States, including Alaska (Figure 1).

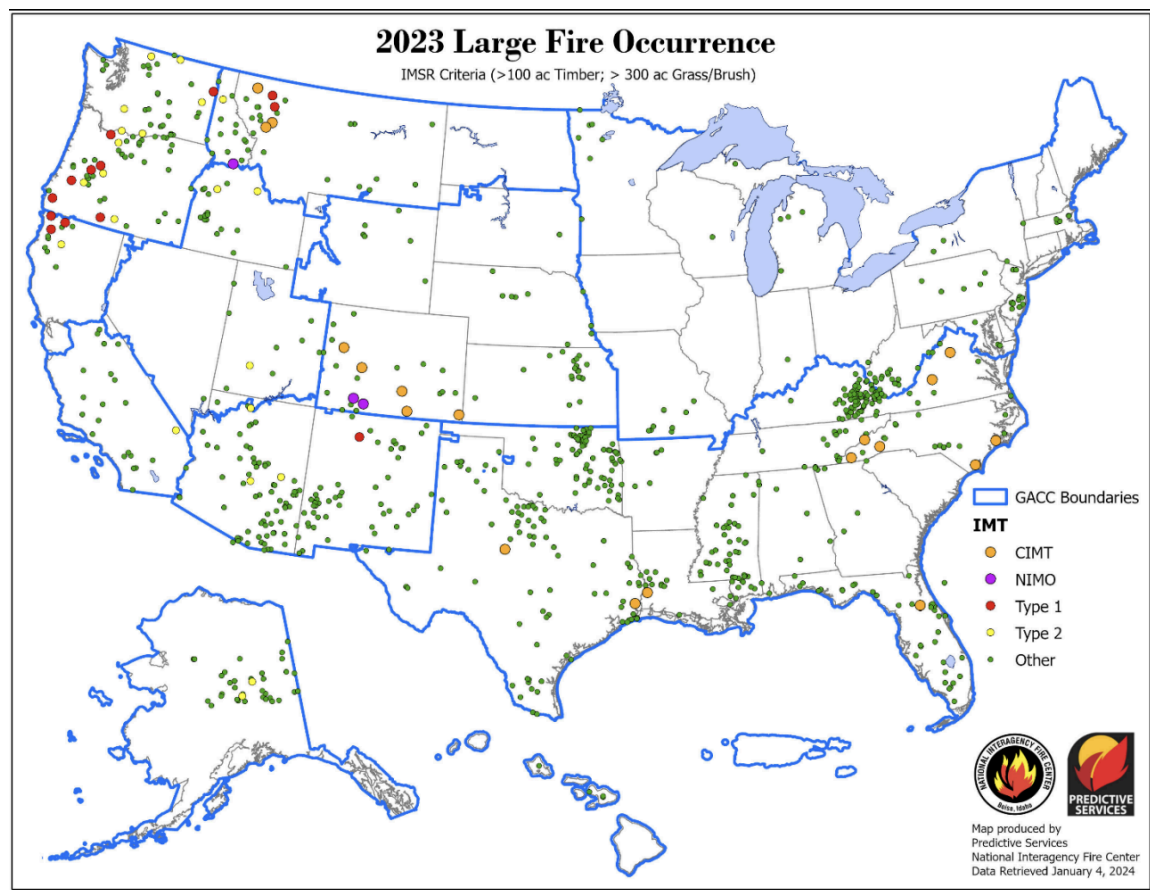


Figure 1. Map of wildfires in 2023 that burned at least 100 acres of timber and 300 acres of grass [2].

According to a 2023 study by the Joint Economic Committee, wildfires are estimated to cause \$384 to \$894 billion in damages each year [3]. The total was calculated from a variety of different financial sources, including watershed costs, healthcare costs, and real estate costs (Figure 2), annual financial costs from diminished real estate value and property damage, are \$337.5 and \$147 billion respectively (Figure 2). Wildfires also cause economic damage to health; exposure to wildfire smoke accounts for \$202.2 billion annually.

Climate-Exacerbated Wildfires Cost As Much as \$893 Billion Per Year

Top-end Annual Total Costs and Losses (Billions \$)

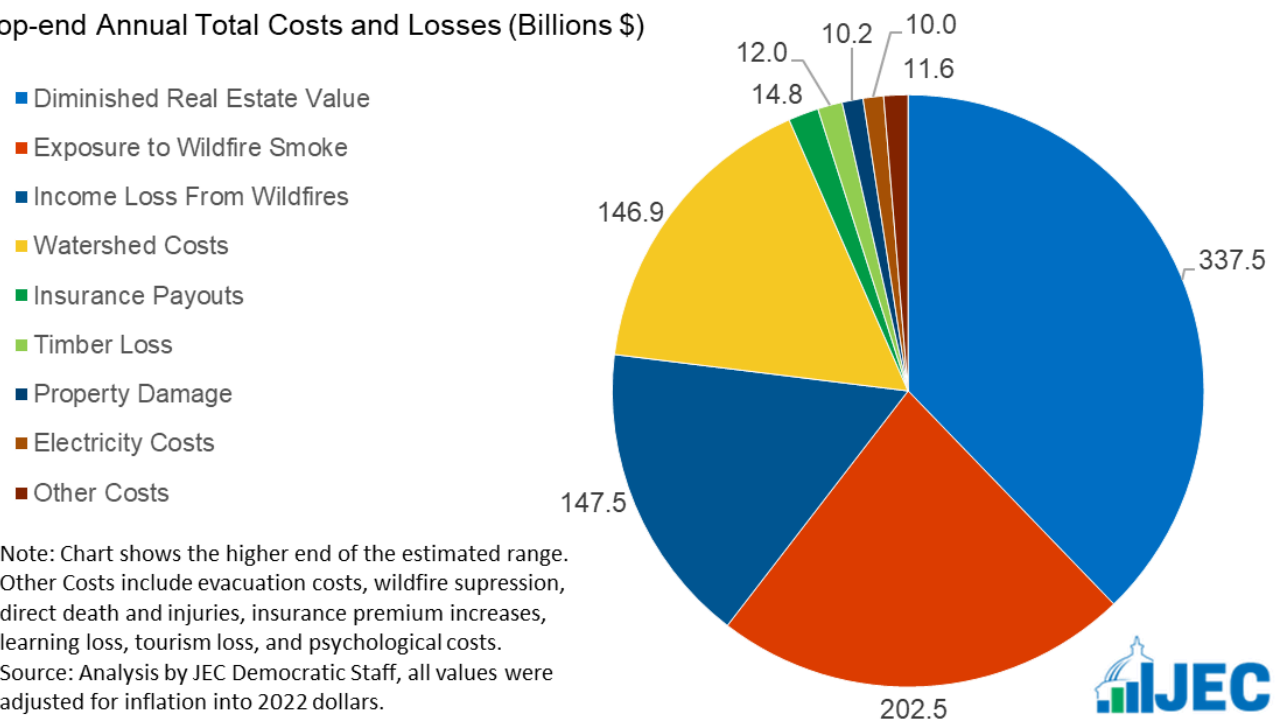
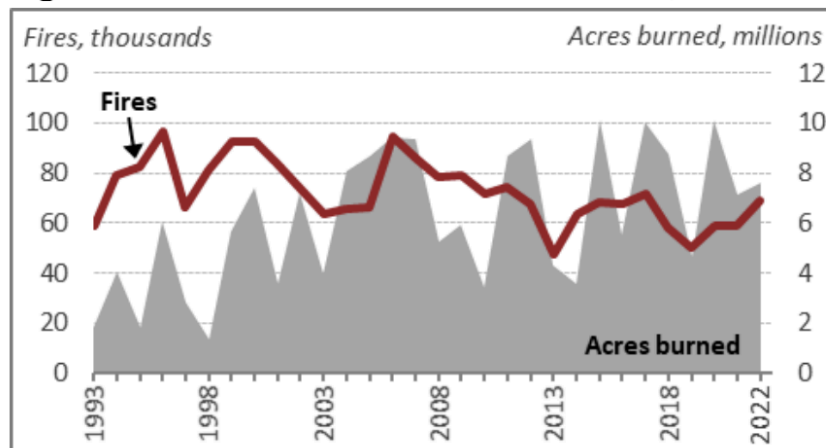


Figure 2. Distribution of economic costs of wildfire in the United States annually [3].

In order to craft a viable solution we have to look at the root causes. Wildfires can have a variety of causes ranging from natural events such as volcanic eruptions or lightning strikes, or human activities such as disposed cigarettes, unattended campfires, arson, or any activities involving the irresponsible usage of fire. This makes wildfires very hard to track since they often stem from small events and aren't noticeable until they begin to spread and substantial damage has been done. In addition, wildfires have become much more intense due to climate change making certain environments hotter and drier conditions, which are ideal for wildfires to form [7]. In this graph from the NICC (Figure 3), it can be observed that wildfires have been burning more and more acres of land despite the average number of fires staying about the same which is concerning. Despite having a variety of tools and systems in place to combat wildfires, detection and prevention systems are lacking in precision and efficiency leading to increased damage.

Figure 1. Annual Wildfires and Acres Burned, 1993-2022

Source: NICC Wildland Fire Summary and Statistics annual reports.

Note: Data reflect wildland fires and acres burned nationwide, including wildland fires on federal and nonfederal lands.

Figure 3. Graph from the NICC (National Interagency Coordination Center) relating the number of wildfires in thousands to the number of acres of land burned in millions [2].

1.2 Aim of Our Solution

Knowing the causes of wildfires, the best way to tackle the problem is to identify wildfires quickly before they get out of hand. This allows nearby residents to evacuate quickly and safely. Although it is rare for people to be killed by wildfires, as mentioned before, exposure to wildfire smoke causes around two hundred billion dollars in healthcare damages annually (Figure 2). This information led us to make early detection a core feature of our device. By allowing people to evacuate quickly, we can reduce the amount of potential health damage. Identifying the fires before they are too big, also makes it more manageable to contain them, preventing further harm to properties and the environment. Our device also covers a variety of indicators to ensure nothing potentially alarming goes undetected by including multiple types of sensors. Any delay in the detection of wildfires allows the fire to grow substantially, therefore increasing the difficulty of extinguishing them.

1.3 Current Solutions

The current state of wildfire detection relies heavily on satellite imagery and either manned or automated watchtowers. One example of satellite imagery is NASA's Fire Information for Resource Management System (FIRMS), which gives live images and updates of a fire's location and growth [4]. Figure 4 shows a visual of the 2020 wildfires in Córdoba, Argentina, created by the NASA FIRMS system.



Figure 4. Satellite image of wildfires in Córdoba, Argentina, in 2020 by NASA [4]
(Red areas indicate wildfire hotspots)

The FIRMS system is very effective at tracking the spread of a fire once it has been discovered, but it lacks the ability to detect a potential fire before there is widespread smoke or flames. A similar issue arises with the watchtowers, where a visual trigger is needed for the fire to be detected [5]. Once there is a large flame or smoke, the fire has often spread too much to be easily contained by firefighters. Additionally, there are current models for predicting the risk of fires, but they rely on weather reports from the general area, and lack measurements of other important factors such as soil moisture levels in specific areas [6]. These risk reports are general for an area and are not frequently updated based on real-time data from the location.

When designing a solution to this issue, a number of challenges must be considered, including the harsh weather conditions that the sensor will have in nature, including heat resistivity in case a fire starts in the area. Additionally, there must be a constant power supply to the device and a security system to limit third-party access to the system.

1.4 Our Proposal

The proposed smart detection system for wildfire monitoring aims to utilize a network of sensor-equipped nodes to gather key environmental data, such as wind speed, temperature, humidity, soil moisture, and VOC gasses from drying vegetation—each a critical indicator of potential wildfire risks [8]. These nodes are designed to be independently powered via solar energy, allowing for sustainable, long-term operation even in remote areas. These nodes lie within the interior of the device and openings are placed in specific locations on the case to allow for accurate data collection. For example, an opening can be found on both the left side and bottom of the device, respectively allowing the soil sensor to reach the ground, and allowing the wind sensor to reach the air above.

The main advantage of our device is that it can be strategically placed and spread out in fire-prone locations so that it can immediately relay information rather than relying on slower systems to detect and respond to wildfires. We also added a multitude of intuitive features to allow the device to be as low maintenance as possible. For example, the addition of closable vent doors allows our device to minimize the amount of debris that enters the vents, in turn preventing the need for regular cleaning or recalibration. The exterior of our device is also plated with fiberglass-infused plastic in order to protect it from environmental factors such as fires, debris, and wildlife.

We decided that the primary users for our smart detection device would be large organizations or the government due to the device's cost and its need for regular maintenance. In addition, our device is made for use by professionals and allows government agencies such as public safety and ecosystem management to gather accurate and helpful data to protect both people and the environment. We knew that government agencies would have the budget to acquire our devices because of the positive tradeoff of the increased safety of the public, but our device was not made to be available at the commercial level, but if public demand reaches a reasonable peak, the device could be adjusted to be more consumer-friendly by decreasing the size, reducing the amount of maintenance needed, and simplifying the functionality. Finally, the device would need to have its costs reduced either by replacing the most expensive components with more cost-efficient ones, or changing the materials used to make the device more appealing and affordable to consumers.

The end goal of our device is to have it be widespread so that it is always in the proximity of potential wildfires due to strategic placement and covers an adequate area of land per device. With our device being used in the field people can have more peace of mind knowing that detection systems are in place on the ground instead of relying on satellite imagery which can be unclear or untimely. In addition to being a detection device, our design gives the device a secondary function as a data collection tool. Since our device has multiple sensors active at all times, it allows scientists and researchers to gather various forms of data about wildfires which can be used to determine wildfire frequency, intensity, or potential catalysts/warning signs.

2. Project and Solution Concept

With the goal of reducing wildfire detection speed and providing data for better risk assessment models, the primary focus of our design revolves around accurate data collection and quick transmission times. As previously mentioned, wildfires are easier to combat the quicker that they are detected, which is an issue that arises with many of the current solutions to the issue of wildfire detection. The main goals of our project are to design a device that is resistant to weather and fires, self-sustainable, low cost, and with accurate environmental sensors. For the wildfire detection itself, this relies on the temperature sensor. However, additional environmental sensors are necessary for generating wildfire risk models, including humidity, soil moisture, VOC gas, and wind. With these goals in mind, specific requirements for different sections of the design were created. An early version of this design can be seen in Figure 5.

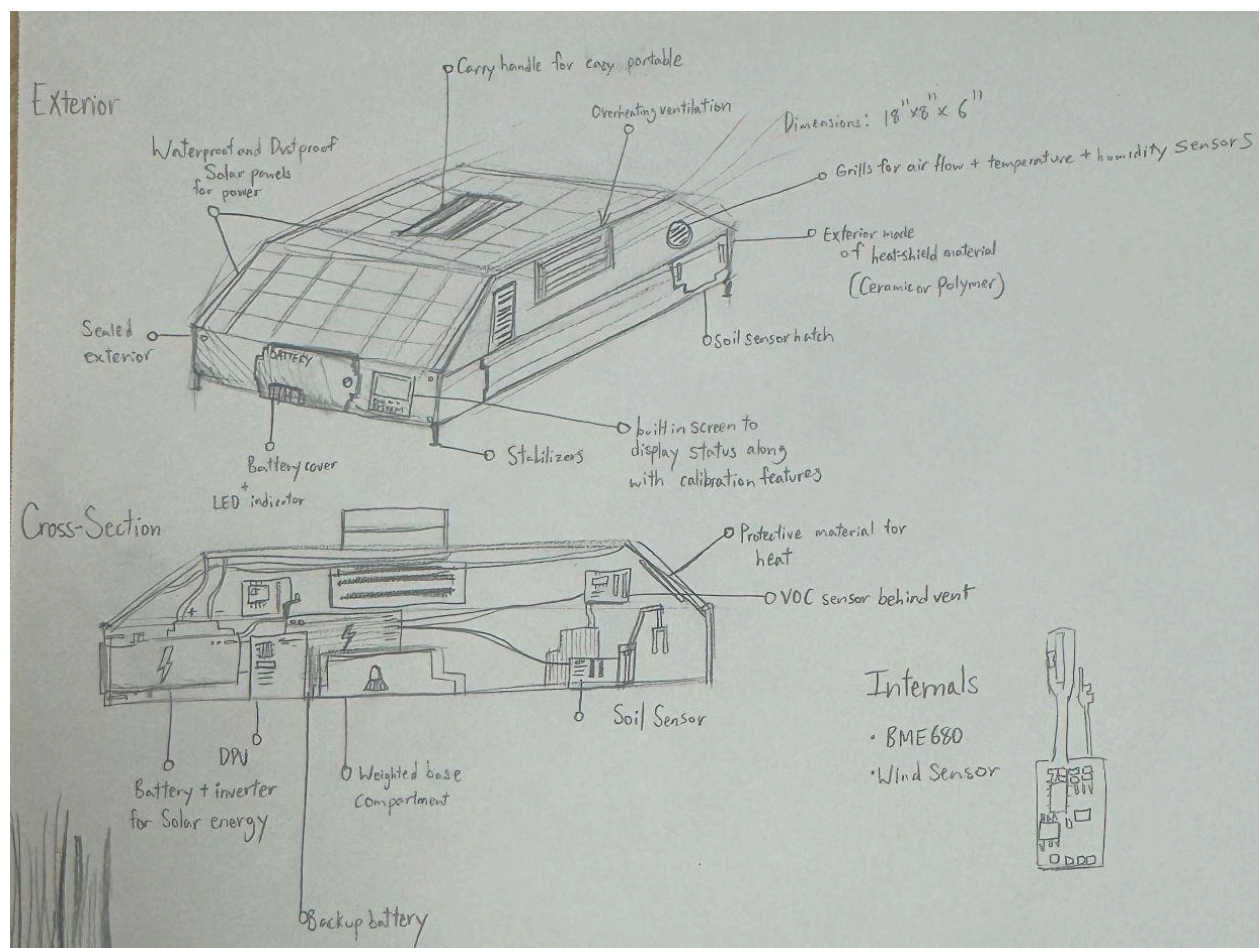


Figure 5. Labeled sketch of sensing device design.

2.1 Hardware Requirements

The first hardware requirement is that the device must be self-sustaining with regard to power. In order for this to be validated, the design must incorporate a solar panel that can provide more energy to the device than it needs to run its functions. Based on the dimensions of our design and

the efficiency of solar panels, it was determined that the solar panel could generate 100 W of power, so the device must run on less than this value. Additionally, the design must also incorporate a battery to store energy when sunlight is in excess in order to maintain power to the device during the night or cloud cover. To prevent the device from being moved by animals or other factors, it requires that the device be above a certain weight threshold, which was decided to be 100 pounds for extra caution. Additionally, the sensing device must be able to withstand various weather conditions along with the high temperature of a forest fire. To validate this requirement, a case material needs to be selected that is waterproof and has a temperature threshold. Wildfires typically burn around 800 degrees Celsius, so the device will need to be able to withstand temperatures above this threshold [6]. Finally, to reduce attention brought to the device, it is preferred that the majority of the case is designed with camouflage or natural colors.

2.2 Performance Requirements

In order to measure the success of our design from a performance perspective, additional performance requirements have been developed. First, the sensors must provide accurate information, and a threshold must be set for the allowed error for these environmental sensors, which was decided to be 5% when compared to a known value. With the primary goal of this project being to increase the detection speed of wildfires, the speed of the temperature is extremely important. It was decided that the temperature sensor refresh rate must be a maximum of every 10 minutes, but preferably much lower, allowing for a quick detection speed when compared to current solutions. Finally, the data must be transmitted to sites for processing, so the sensor must be able to transmit a minimum distance of 15 km before the signal can be picked up by a tower that can send it to nearly any location.

2.3 Budget Requirement

With the majority of the current wildfire detection technology being satellite-based, it is hard to develop an idea for the cost of a sensing device like our design. However, there are drone-based detection systems in use, with drones typically costing approximately \$4000, so it is a requirement for our solution to cost less than \$4000, preferably by a wide margin.

3. Technical Approaches

3.1 Hardware Validation

Five hardware requirements needed to be validated for this project: A maximum power consumption of 100 watts, a weight greater than 100 pounds, water resistance, fire resistance, and camouflage. Our design's power consumption was calculated by summing the individual power consumption of each component. For our sensors, the manufacturer did not state how

much power was required. So we calculated the power consumption for the sensors by using the formula:

$$W = V * I$$

Where W is wattage, a unit for power, V is voltage, and I is current. For the other components, we obtained the power consumption from their information section. Weight was also calculated by summing the individual values of each component.

Water resistance was not mathematically validated. Instead, we aimed to pick a water-resistant case material, and mechanism that stops any water from getting into the device. Fire resistance was validated by the characteristics of the case material we chose. Specifically, the requirement was that the material was known to insulate things from temperatures up to 1200 °C [9].

3.2 Performance Validation

Three performance requirements needed to be validated for this project: a maximum sensor error of +/- five percent, a data relay range of 15 kilometers, and a wildfire detection speed of under 10 minutes.

Validating the sensor accuracy was simple. In the information section, each sensor manufacturer stated the accuracy of the sensor in percentage points. The greatest error threshold of any of our chosen sensors was +/- three percentage points, which is within our requirements. Validating the data relay range was also simple. In the information section, antenna manufacturers state the range of the antenna in kilometers. The antenna in our design has a range of 20 kilometers which surpasses our requirement by 33 percentage points.

Validating the detection speed of wildfire was slightly more tricky. To simplify things, we made a couple of assumptions. We assumed that a wildfire could be detected simply by checking that the temperature was above a certain value. Therefore the requirement of 10 minutes can be validated by the refresh rate of the temperature sensor.

3.3 Budget Validation

To validate our budget requirement, we simply summed the individual prices of each component in our design.

4. Results & Discussion

In completion of our project's design and components, we successfully validated our hardware, performance, and budget requirements while creating the device's schematic and sketch. These results provide a proof of principle demonstrating that our device conceptually works for early wildfire detection and sets up grounds for future development and improvements.

Req #	Req. Name	Priority	Pass/Fail Criteria	Design Outcome	Validation Method
1. Hardware Requirements					
1.1	Solar Power	1	100 W	73.05 W	Calculated
1.2	Weight	2	>100 lb	130 lb	Calculated
1.3	Water Resistance	1		PASS	Case Material
1.4	Fire Resistance	1	800 C	1200 C	Case Material
1.5	Camouflage	2		PASS	Case Design
2. Performance					
2.1	Sensor Accuracy [Wind, VOC, Soil Humidity, Air Humidity]	1	$\pm 5\%$ Accuracy	Worst: $<\pm 3\%$ Accuracy	Material Description
2.2	Range of Data Relaying	1	15 km	20 km	Antenna Range
2.3	Wildfire Detection Speed	1	<10 minutes	1 minute	Temperature Sensor Refresh Rate
3. Cost					
3.1	Price	1	\$4,000	\$1,150	Calculated

Table 1: Requirements and Validation Table (green represents Pass).

4.1 Hardware Validation: Can it meet the environmental and operational demands?

The hardware validation results are summarized in Table 1. As shown, the design successfully meets all hardware requirements, including power consumption, weight, water resistance, fire resistance, and camouflage. Notably, the fire resistance of the selected case material, fiberglass-infused plastic, exceeds the required 800°C by 400°C.

4.2 Performance Validation: Does it accurately detect and relay information timely?

The performance validation (Table 1), demonstrated that the system meets all criteria. The sensory accuracy was verified to have a worst-case error of $\pm 3\%$, well within the $\pm 5\%$ threshold. The data relay range of 20 km surpasses the minimum requirement of 15 km, and the wildfire detection speed is validated to respond in 1 minute, significantly faster than the required 10 minutes.

4.3 Cost

The total design was calculated at \$1,150, well below the \$4,000 budget threshold.

4.4 Schematic and Sketch

To illustrate the proposed design and its functionality, a schematic and conceptual sketch were developed. These provide a visual representation of the system's components and their integration, serving as a basis for understanding the design's functionality and meeting the project's requirements.

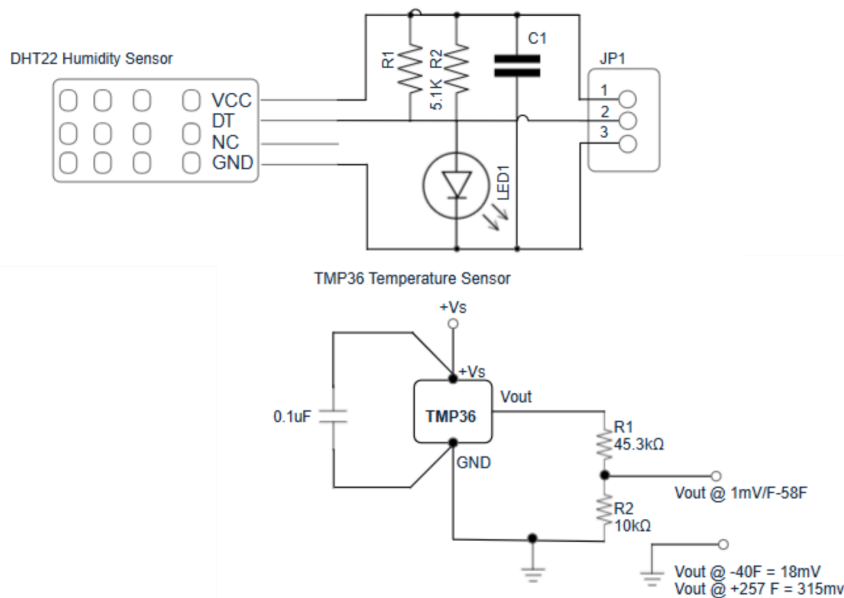


Figure 6: DHT22 & TMP 36 Schematic

This portion of the schematic (Figure 6) represents the DHT22 (Humidity) and TMP36 (Temperature) sensors. The DHT22 sensor provides digital humidity and temperature readings while the TMP36 provides an analog output voltage corresponding to the temperature. These sensors are confined to the power requirements as well (<100 watts). This schematic (Figure 5) outlines the electrical configuration of the sensor components, showing how power is distributed

across the system. This was essential in verifying that the system could operate within the specified power consumption limits and support data relay.

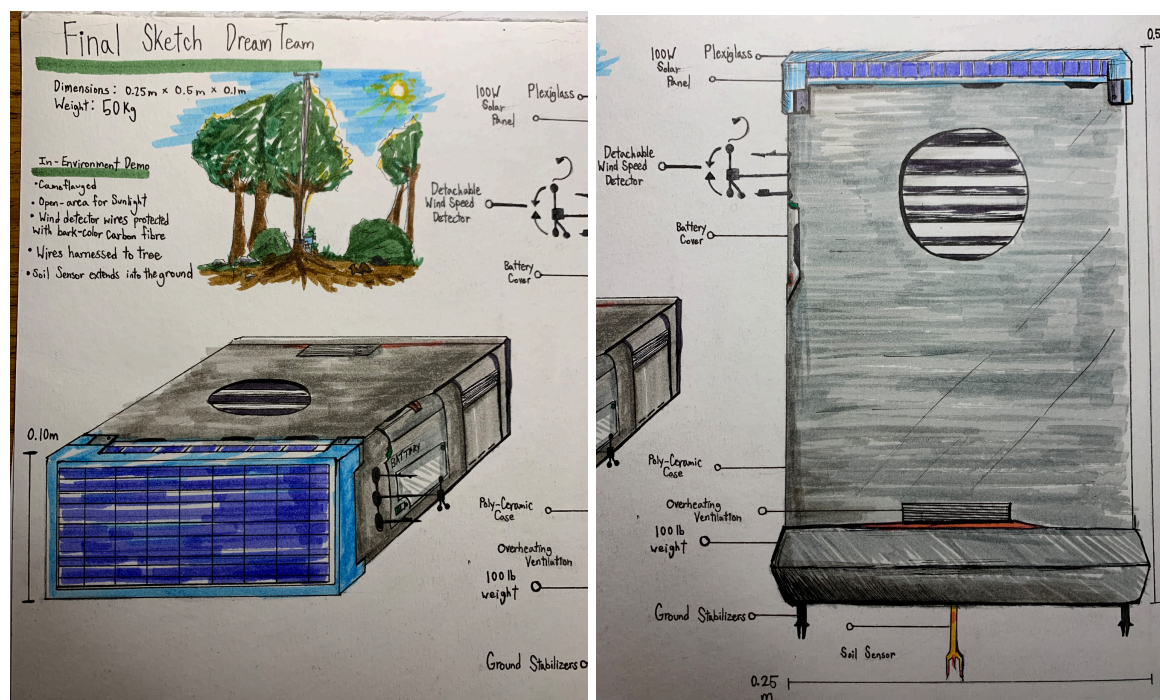


Figure 7 & 8: The final sketch of the exterior design and its use in the environment.

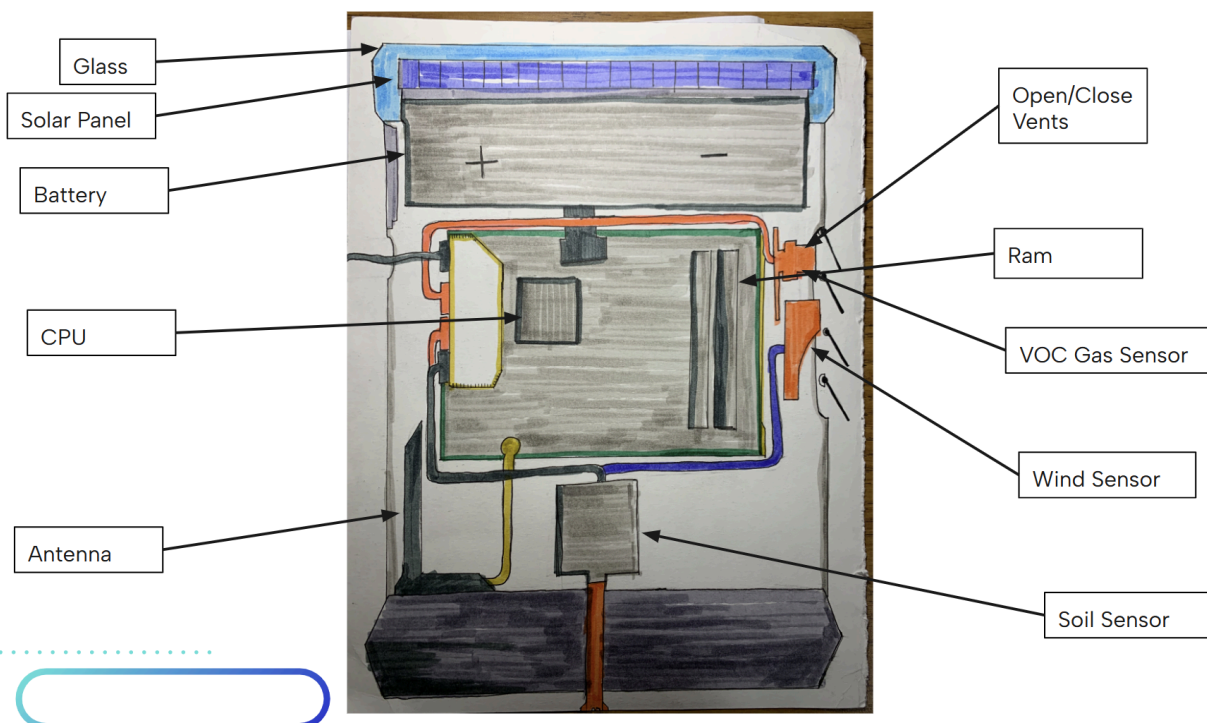


Figure 9: The internal components of our design.

Figure 7 and Figure 8 provide a detailed visualization of the device's dimensions and external components. The overall dimensions make it compact and transportable. The front view

highlights key features such as the solar panel which powers the device, vents which open to allow for air humidity measurements and close to prevent water from entering the device during storms. The internal components (Figure 9) of the device include several key sensors and hardware essential for monitoring environmental conditions. These components are organized within the device's housing: the VOC gas sensor detects VOC gases, the wind sensor tracks wind speed and direction, and the soil sensor measures soil moisture and pH. The CPU processes the data, and the RAM stores it for later use. The antenna allows for wireless data transmission. This internal Setup ensures that the device can efficiently collect and transmit environmental data.

4.5 Next Steps

This section outlined the design and validation of our environmental monitoring device, detailing both the hardware and performance requirements, as well as the schematic and sketch of the system. The device meets key specifications such as power consumption, sensor accuracy, and operational range, with the visual representations clearly illustrating the integration of external and internal components.

The device has a significant social impact, benefiting industries such as agriculture, environmental monitoring, and safety. It can help farmers monitor soil and weather conditions, aid environmental agencies in pollution tracking, and assist in early wildfire detection, ultimately improving public health and safety.

Looking ahead, future research and development should focus on improving battery life, reducing costs, and enhancing sensor accuracy and data relay range. With more time, we could explore better energy storage solutions, lower production costs, and incorporate advanced features like real-time data analytics. For those attempting similar projects, understanding these limitations is crucial, and addressing them through efficient design and technology improvements would be key to creating a more impactful solution.

Conclusion

To conclude our project, we want to restate the importance of the prevention of future wildfires as they are greatly detrimental to both humankind and the environment. Although many different systems for wildfire prevention currently exist, wildfires undeniably still cause a significant amount of damage to everything in their path. Through our research, we noticed the lack of an efficient ground-based detection system for wildfires and wanted to implement a system to tackle that weak spot in the sector of wildfire prevention. This led us to develop a unique device to allow people to reliably detect, predict, and ultimately prevent these devastating fires right as they begin. We eventually decided that the core objectives for the project were to create a device that requires low maintenance, is both energy efficient and cost-effective, and was able to relay accurate and useful information to fight wildfires.

We determined that a ground-based system design would be the most realistic and beneficial to the realm of wildfire prevention by covering a weak point of current solutions while complementing existing systems. By filling in for the lack of ground-based detection systems, wildfire prevention as a whole would improve since it would now have a more well-rounded approach. Through our results and calculations, we proved the efficiency and effectiveness of our device and that it would aid us in completing the objectives of providing a quick response to wildfires and relaying environmental data. We were ultimately able to design a device that met all our objectives and validate that it would be able to withstand harsh environmental conditions, have a total cost of only around 1150 US dollars, have a detection speed of less than 10 minutes, and have an effective range of 20 km. This deduced that a ground-based device like ours would be affordable and efficient to implement. The placement of our device throughout high-risk areas allows for prediction capabilities far quicker than the current measures, giving those in fire-prone areas peace of mind and increased security. In conclusion, our device provides a powerful and compact asset to the sector of wildfire prevention and simultaneously enables further research through consistent data collection.

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Appendixes

Name	Contributions
Ethan Connolly	Project and Solution Concept section, paragraphs 1, 2, and 3 of the executive summary, initially wrote current solutions part of the Introduction, formatting
Johan Houdebine	Results and discussion, Schematic design, formatting
Steve Tchana	Technical Approaches, Final Sketch, grammatical revisions, formatting
Aaron Zhang	Prototype sketch, Final Sketch, 1.1, 1.2, 1.4 of Introduction, Conclusion, paragraphs 4, 5 of the executive summary, references formatting